Diffractive Deep Neural Networks: Physics-Informed Initialization and Robustness

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Introduction

Motivation: The growing computational demands of neural networks motivate alternative computing architectures.

Optical computing:

- Information processing through the physical propagation of light.
- Intrinsic parallelism and the potential for energy-efficient inference.

Diffractive deep neural networks (D2NNs): Classification through the diffraction of coherent light by a sequence of phase masks. The phase profiles are optimized by gradient descent.

Two research questions:

1. How can physical structure guide the initialization of phase masks?
2. How sensitive are trained networks to wavelength shifts and fabrication errors?

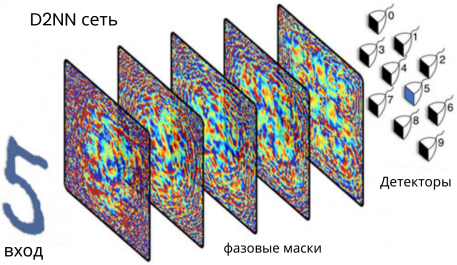
D2NN Architecture

A D2NN consists of:

- N planar phase masks;
- free-space propagation between masks;
- a detector measuring the output intensity distribution.

Operations in each layer:

1. Phase modulation.
2. Free-space diffractive propagation.



Optical setup

Physical Model

Each phase mask m is described by a phase profile $\varphi_m(x, y)$ and applies the transformation

$$A_m^+(x, y) = A_m(x, y)\, e^{i\varphi_m(x,y)}.$$

Here, A_m and A_m^+ are the fields immediately before and after the mask.

- The trainable phase profiles control the interference pattern.
- Free-space propagation couples different spatial locations.
- For fixed masks, modulation and propagation are linear in the complex field.
- Intensity detection introduces the quadratic mapping $I = |A|^2$.

Helmholtz Wave Equation

A monochromatic field is represented by its complex amplitude:

$$E(\mathbf{r}, t) = \Re\{A(x, y, z)e^{-i\omega t}\}.$$

In a homogeneous medium, the amplitude satisfies the Helmholtz equation:

$$\nabla^2 A + k^2 A = 0, \qquad k = \frac{2\pi n}{\lambda}.$$

The solution is a superposition of plane waves. For a transverse component $e^{i(k_x x + k_y y)}$, the longitudinal wave number is

$$k_z = \sqrt{k^2 - k_x^2 - k_y^2}.$$

Angular Spectrum Propagation

Step 1. Fourier transform in the input plane:

$$\widetilde{A}(k_x, k_y; 0) = \iint A(x, y, 0) e^{-i(k_x x + k_y y)}\, dx\, dy.$$

Step 2. Propagation over a distance z:

$$\widetilde{A}(k_x, k_y; z) = \widetilde{A}(k_x, k_y; 0) e^{ik_z z}.$$

Step 3. Inverse Fourier transform:

$$A(x, y, z) = \frac{1}{(2\pi)^2} \iint \widetilde{A}(k_x, k_y; z) e^{i(k_x x + k_y y)}\, dk_x\, dk_y.$$

Implemented numerically using the fast Fourier transform (FFT).

D2NN: Mathematical Formulation

Let $\mathcal{P}_d$ denote angular spectrum propagation over a distance d and $\mathcal{M}_{\varphi_m}$ denote phase modulation:

$$\mathcal{M}_{\varphi_m}[A] = A\, e^{i\varphi_m}.$$

For N masks and $N+1$ propagation gaps, the output field is

$$A_{\text{out}} = \mathcal{P}_{d_{N+1}} \circ \mathcal{M}_{\varphi_N} \circ \mathcal{P}_{d_N} \circ \cdots \circ \mathcal{M}_{\varphi_1} \circ \mathcal{P}_{d_1}[A_{\text{in}}].$$

Output signal:

$$I(x, y) = |A_{\text{out}}(x, y)|^2.$$

The detector converts regional intensity measurements into classification logits.

Trainable parameters: phase profiles $\{\varphi_1, \ldots, \varphi_N\}$.
Fixed parameters: propagation distances and optical geometry.

Training

Cross-entropy loss:

$$\mathcal{L} = -\sum_{k=1}^{C} y_k \log p_k,$$

One-hot target: y; predicted probabilities: p.

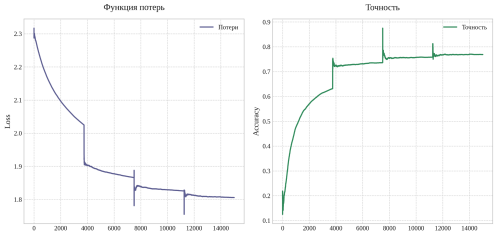
Training pipeline:

1. Average intensity over each class region R_k:

$$\ell_k = \text{mean}_{(x,y)\in R_k} |A_{\text{out}}(x,y)|^2.$$

2. Apply softmax to obtain p_k.
3. Update the phase profiles using Adam.

Settings: learning rate 10^{-3}; weight decay 10^{-6}; evaluation by classification accuracy.



Validation loss and accuracy

Physics-Informed Initialization: Gerchberg–Saxton

Motivation: Replace random or Xavier initialization with a physically meaningful starting point.

The Gerchberg–Saxton (GS) algorithm alternates between the input and target amplitude constraints while updating the phase.

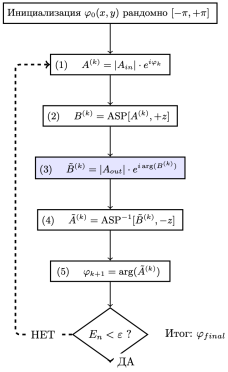
Focusing error:

$$E_k = \left\| \, |B^{(k)}| - |A_{\text{target}}| \, \right\|_2^2.$$

Gaussian target amplitude:

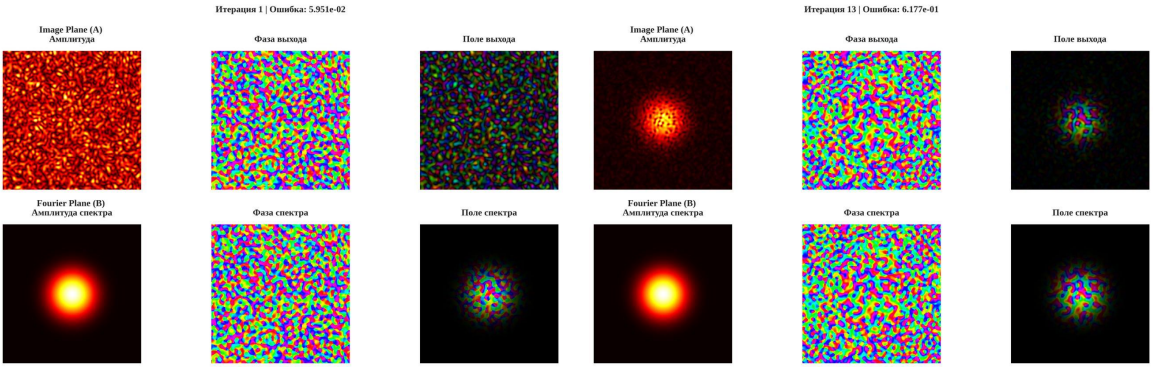
$$A_{\text{target}}(x, y) = \exp\left(-\frac{x^2 + y^2}{2\sigma^2} \right).$$

This target concentrates light near the center of the output plane.



Gerchberg–Saxton iteration scheme

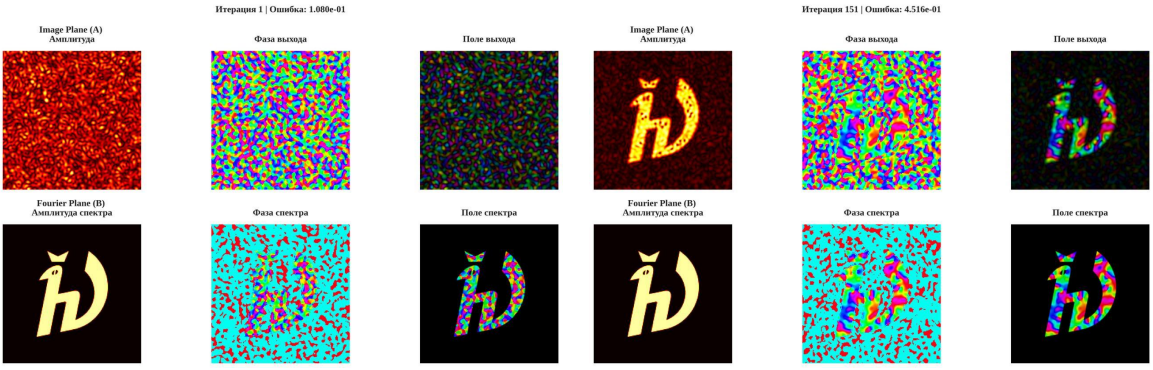
Gerchberg–Saxton Iterations: Gaussian Target



Iteration 1

Iteration 13

Gerchberg–Saxton Iterations: Structured Target



Iteration 1

Iteration 151

Gerchberg–Saxton: Non-Increasing Focusing Error (1/2)

Assumption. The propagation operator P is unitary: only propagating components are retained, with no aperture loss or evanescent attenuation. Since $|e^{ik_z z}| = 1$, Parseval's theorem gives

$$\|PX - PY\|_2 = \|X - Y\|_2.$$

Claim. $E_{k+1} \le E_k$, where $E_k = \sum_n \big(|B^{(k)}[n]| - r_n\big)^2$ and $r_n = |A_{\text{target}}[n]|$.

Step 1. Projection onto the target amplitude constraint. The closest point to $B^{(k)}$ in $\mathcal{C}_2 = \{B : |B[n]| = r_n\}$ is

$$\widetilde{B}^{(k)}[n] = r_n e^{i \arg B^{(k)}[n]}.$$

Indeed,

$$|B^{(k)}[n] - r_n e^{i\theta_n}|^2 = |B^{(k)}[n]|^2 + r_n^2 - 2r_n |B^{(k)}[n]| \cos\big(\theta_n - \arg B^{(k)}[n]\big)$$

is minimized by matching the phase. Thus $\|B^{(k)} - \widetilde{B}^{(k)}\|_2^2 = E_k$.

Gerchberg–Saxton: Non-Increasing Focusing Error (2/2)

Step 2. Back-propagation and the input amplitude constraint. Define $\widetilde{A}^{(k)} = P^{-1}\widetilde{B}^{(k)}$ and project it onto $\mathcal{C}_1 = \{A : |A[n]| = |A_{\text{in}}[n]|\}$ to obtain $A^{(k+1)}$. Since $A^{(k)} \in \mathcal{C}_1$,

$$\|A^{(k+1)} - \widetilde{A}^{(k)}\|_2 \leq \|A^{(k)} - \widetilde{A}^{(k)}\|_2.$$

Step 3. Combine both projections with unitary propagation. For $B^{(k)} = PA^{(k)}$,

$$\begin{aligned} E_{k+1} &\leq \|B^{(k+1)} - \widetilde{B}^{(k)}\|_2^2 \\ &= \|A^{(k+1)} - \widetilde{A}^{(k)}\|_2^2 \\ &\leq \|A^{(k)} - \widetilde{A}^{(k)}\|_2^2 \\ &= \|B^{(k)} - \widetilde{B}^{(k)}\|_2^2 = E_k. \end{aligned}$$

Scope: The focusing error is non-increasing under these assumptions. This does not guarantee a global optimum or monotonic decrease of the subsequent classification loss.

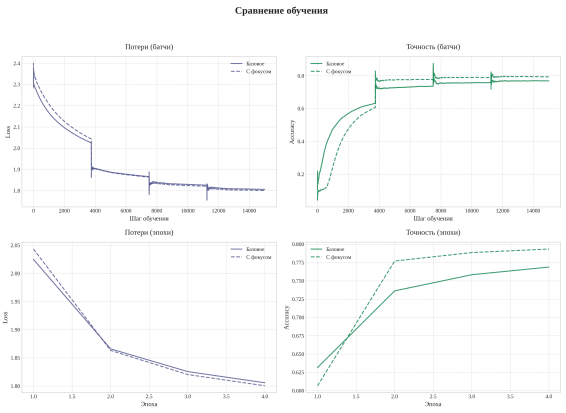
Effect of Initialization on Training

Two initialization strategies:

1. **Random:** phase profiles initialized without a focusing objective.
2. **GS initialization:** phase masks optimized to act as focusing elements before classification training.

Reported MNIST comparison:

- GS initialization: **83.55%** final accuracy.
- Random initialization: **82.95%**.
- Similar convergence speed in the presented experiment.



Training comparison

Phase Masks: From Phase to Thickness

The learned phase profile can be converted into a thickness profile:

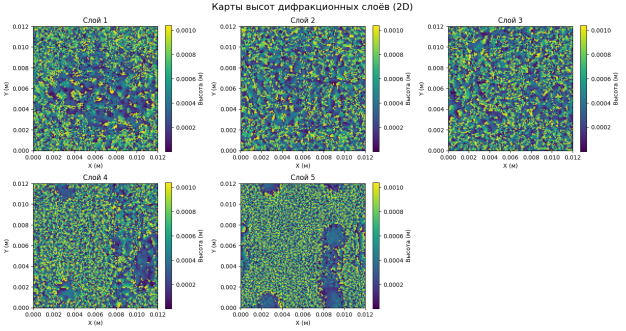
$$h(x, y) = \frac{\lambda\, \Phi(x, y)}{2\pi\, \Delta n},$$

where $\Delta n = n_{\mathrm{mat}} - n_{\mathrm{air}}$.

Optical parameters:

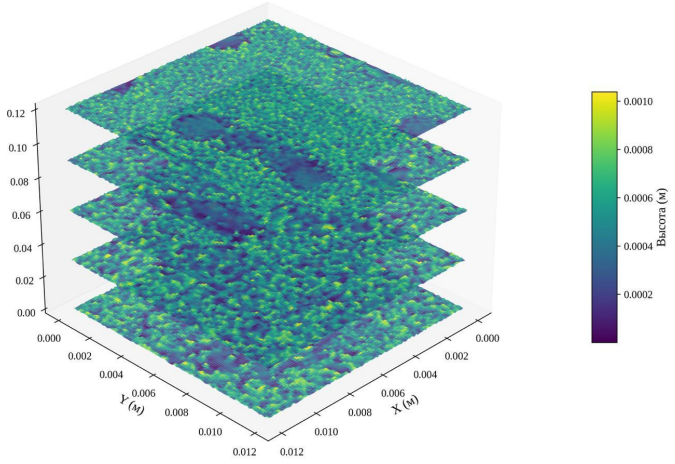
Wavelength	$\lambda = 0.75\,\mathrm{mm}$
Material index	$n_{\mathrm{mat}} = 1.7227$
Air index	$n_{\mathrm{air}} = 1.0003$

Material: VeroBlackPlus RGD875 photopolymer.



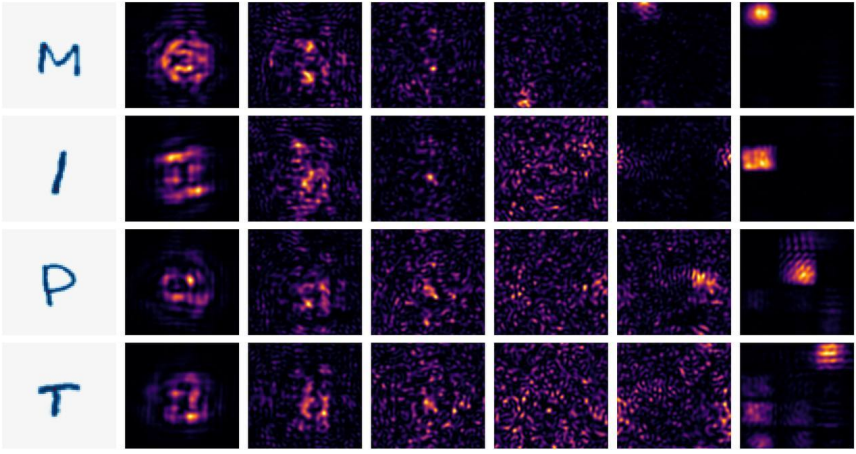
Phase profiles of the individual layers

Phase Masks: 3D Visualization



Simulated thickness profiles of the D2NN phase masks

Intensity Evolution Along the Optical Axis



Intensity distribution $I(x, y)$ after successive layers

Robustness: Computational Experiments

Phase encoding:

$$A_{\text{in}}(x, y) = e^{i\phi(x,y)}.$$

Amplitude encoding:

$$A_{\text{in}}(x, y) = a(x, y).$$

For each encoding, compare:

- standard training;
- robust training.

Evaluation criteria:

- Classification accuracy.
- Sensitivity to wavelength shifts.
- Sensitivity to phase-mask perturbations.
- Effect of training noise on robustness.

Goal: Compare accuracy and physical robustness across input encodings and training regimes.

Robust Training

Random perturbations are added to the phase profiles during training to model fabrication errors.

Perturbation model:

$$\widetilde{\varphi}(x,y) = \mathrm{mod}\big(\varphi(x,y) + \delta(x,y),\, 2\pi\big),$$

$$\delta(x,y) \sim \mathcal{N}\big(0, \sigma^2(x,y)\big), \qquad \sigma(x,y) = \alpha\,|\varphi(x,y)|.$$

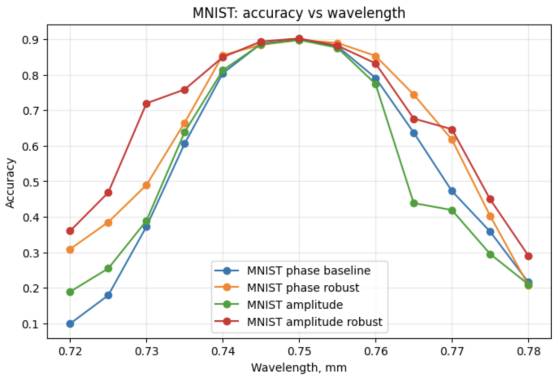
The parameter α controls the relative noise level during training.

Purpose: Reduce the accuracy loss caused by imperfections in the phase masks.

Robustness to Wavelength Shifts

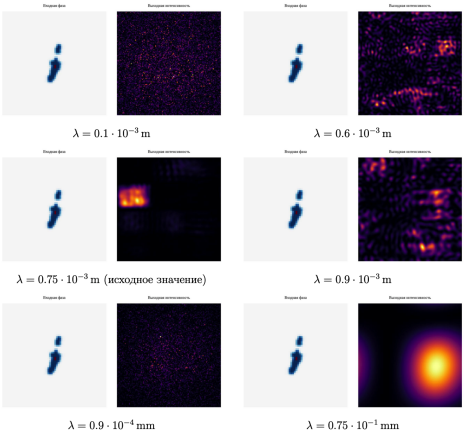
- The highest accuracy is observed near $\lambda \approx 0.74$–$0.76\,\mathrm{mm}$.
- Accuracy decreases away from the design wavelength.
- Robust training broadens the usable wavelength range in the reported experiments.

Observation: Robustly trained models are less sensitive to deviations from the nominal wavelength.



Classification accuracy versus wavelength

Wavelength Analysis: Detector Intensity



Intensity distributions at the detector plane

Robustness to Fabrication Noise

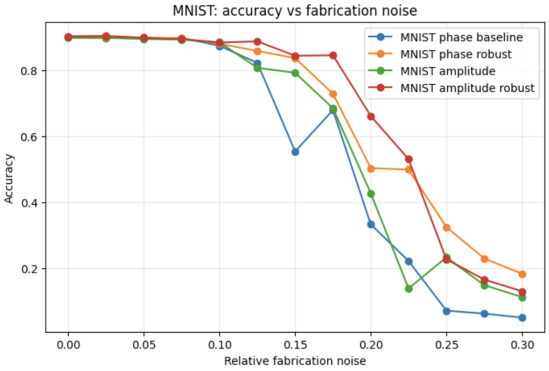
Test-time perturbation model:

$$\delta(x, y) \sim \mathcal{N}\big(0, \sigma^2(x, y)\big),$$

$$\sigma(x, y) = \eta\,|\varphi(x, y)|.$$

Here, η is the relative fabrication-noise level used at evaluation.

Result: With robust training, accuracy degrades more slowly as fabrication noise increases.



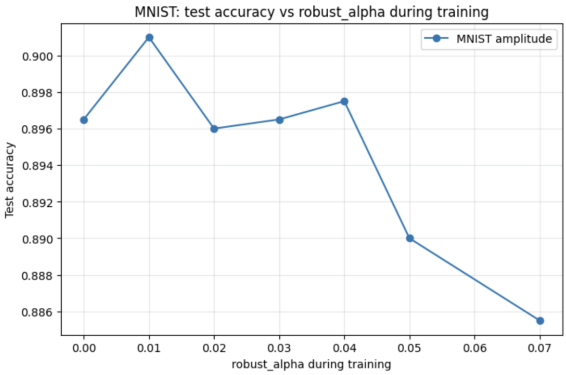
Classification accuracy versus fabrication-noise level

Effect of the Training Noise Level

For each input encoding, models were trained with different values of the noise parameter α.

- Nominal accuracy changes only slightly for $\alpha = 0$–0.04.
- Larger values, $\alpha \geq 0.05$, reduce accuracy in the reported sweep.

Takeaway: The noise level must balance tolerance to perturbations against nominal classification accuracy.



Classification accuracy versus training noise α

Conclusions

Physics-informed initialization:

- GS provides a physically meaningful alternative to random phase initialization.
- The focusing error is non-increasing under unitary propagation.
- In the reported MNIST comparison, final accuracy was 83.55% with GS versus 82.95% with random initialization.

Robustness:

- Training with phase perturbations reduces sensitivity to wavelength shifts and fabrication noise in the studied settings.
- Excessive training noise reduces nominal accuracy.
- Phase and amplitude input encodings yield comparable accuracy on the studied classification task.

These numerical studies inform the initialization and physical tolerance design of diffractive optical networks.

Thank you!

Kseniia Ibragimova, Rodion Akinzhala